Experiment Title: Package Management in Linux (Debian/Ubuntu)

Topic: Managing Software Packages with APT (`update`, `upgrade`, `install`, `remove`, `autoremove`, `purge`)

1. Objective

To learn software package management in Linux using APT:

- Refresh package lists (**`sudo apt update`**) and upgrade packages (**`sudo apt upgrade`**).
- Install software packages (**`sudo apt install`**).
- Compare removing software (**`remove`**) vs completely deleting settings (**`purge`**).
- Remove unused dependencies (**`sudo apt autoremove`**).

2. Theoretical Background

Linux uses package managers like **`apt`** to fetch, install, update, and remove software packages from official repositories.

2.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`update`	`sudo apt update`	Refresh package list from repositories.	`sudo apt update`
`upgrade`	`sudo apt upgrade`	Upgrade installed packages to newest versions.	`sudo apt upgrade`
`install`	`sudo apt install <pkg>`	Download and install package and dependencies.	`sudo apt install tree`
`remove`	`sudo apt remove <pkg>`	Uninstall package but keep configuration files.	`sudo apt remove tree`
`autoremove`	`sudo apt autoremove`	Remove unused leftover dependencies.	`sudo apt autoremove`
`purge`	`sudo apt purge <pkg>`	Completely delete package and configuration files.	`sudo apt purge htop`

2.2 Comparing Uninstallation Methods: Remove vs. Purge vs. Autoremove

Action	What It Deletes	When To Use
`apt remove`	Application binaries only.	Keep settings for future reinstallation.
`apt purge`	Binaries and configuration files.	Completely erase software and settings.
`apt autoremove`	Unused leftover dependencies.	Clean disk space after removing programs.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul` (with sudo privileges)	**Command Shell**	GNU Bash 5.2
Package Manager	APT (Advanced Package Tool)	**Demo Packages**	`htop`, `tree`

4. Procedure / Steps

1. Update package repository lists using **`sudo apt update`**.
2. Simulate system package upgrade using **`sudo apt upgrade`**.
3. Install utilities **`htop`** and **`tree`** using **`sudo apt install -y htop tree`**.
4. Verify installed paths using **`which htop tree`**.
5. Remove **`tree`** package using **`sudo apt remove -y tree`**.
6. Remove unused dependencies using **`sudo apt autoremove -y`**.
7. Purge **`htop`** and all config files using **`sudo apt purge -y htop`**.
8. Verify removal using **`which htop`**.

5. Result / Output Demonstration

Commands run on Ubuntu under user `ratul`:

5.1 Session 1: Updating Package Lists, Upgrading, and Installing Software

ratul@firefly: ~ — bash

```
ratul@firefly:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
All packages are up to date.
ratul@firefly:~$ sudo apt upgrade -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt install -y htop tree
Reading package lists... Done
The following NEW packages will be installed:
  htop tree
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 188 kB of archives.
Unpacking htop (3.3.0-4build1) ...
Unpacking tree (2.1.1-1) ...
Setting up htop (3.3.0-4build1) ...
Setting up tree (2.1.1-1) ...
ratul@firefly:~$ _
```

Explanation of Session 1 Results:

- **`sudo apt update`**: Fetched latest package index from Ubuntu mirrors without installing packages.
- **`sudo apt upgrade`**: Checked for package updates; all packages were already up to date.
- **`sudo apt install -y htop tree`**: Downloaded and installed both packages automatically.

5.2 Session 2: Uninstalling Packages, Cleaning Leftovers, and Full Purging

ratul@firefly: ~ — bash

```
ratul@firefly:~$ which htop tree
/usr/bin/htop
/usr/bin/tree
ratul@firefly:~$ sudo apt remove -y tree
Reading package lists... Done
The following packages will be REMOVED:
  tree
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing tree (2.1.1-1) ...
Processing triggers for man-db (2.12.0-4build1) ...
ratul@firefly:~$ sudo apt autoremove -y
Reading package lists... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ratul@firefly:~$ sudo apt purge -y htop
Reading package lists... Done
The following packages will be REMOVED:
  htop*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
Removing htop (3.3.0-4build1) ...
Purging configuration files for htop (3.3.0-4build1) ...
ratul@firefly:~$ which htop
(no output -- command not found)
ratul@firefly:~$ _
```

Explanation of Session 2 Results:

- **`which htop tree`**: Confirmed binaries were installed in `/usr/bin/`.
- **`sudo apt remove tree`**: Deleted program binary while preserving configuration files.
- **`sudo apt autoremove`**: Cleaned up leftover unused dependencies.
- **`sudo apt purge htop`**: Removed **`htop`** binary and purged all configuration files.
- **`which htop`**: Returned no output, confirming complete removal.

6. Conclusion

Key takeaways:

1. **`apt update`** refreshes repository lists, while **`apt upgrade`** updates installed software.
2. **`apt install`** downloads and installs programs with required dependencies.
3. **`apt remove`** removes binaries, while **`apt purge`** wipes configuration files as well.
4. **`apt autoremove`** cleans orphaned dependencies to free disk space.